HDOC Day 31

Aim

To write a menu-driven C program using a two-dimensional array to calculate the sum of all matrix elements, store each row sum in a one-dimensional array, and display the transpose of the matrix.

Theory

A matrix is represented in C using a two-dimensional array. Nested loops access its elements. The total sum is obtained by adding every element; row sums are calculated separately and stored in an array. The transpose interchanges rows and columns: T[j][i] = A[i][j].

Algorithm

1. Start and read the number of rows (r) and columns (c).

2. Read the r x c matrix elements.

3. Display a menu with choices 1, 2, 3 and 4; read the choice.

4. Choice 1: initialize sum = 0; traverse the matrix and add each element; display sum.

5. Choice 2: for each row, initialize rowSum[i] = 0; add all elements in that row; display each row sum and the row-sum array.

6. Choice 3: print A[j][i] by iterating columns first and rows second, producing the c x r transpose.

7. Choice 4: exit. For any other choice, print an invalid-choice message.

8. Repeat the menu until choice 4; stop.

C Program

```
#include <stdio.h>
int main() {
    int a[20][20], rowSum[20];
    int r, c, i, j, ch, sum;

    printf("Enter the number of rows and columns: ");
    scanf("%d%d", &r, &c);
    if (r < 1 || r > 20 || c < 1 || c > 20) {
        printf("Invalid matrix size.\n");
        return 0;
    }
    printf("Enter the matrix elements:\n");
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);

    do {
        printf("\n1. Sum of all elements\n");
        printf("2. Sum of each row\n");
        printf("3. Transpose\n4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &ch);
        switch (ch) {
            case 1:
                sum = 0;
                for (i = 0; i < r; i++)
                    for (j = 0; j < c; j++)
```

```
                    sum += a[i][j];
                printf("Sum of all elements = %d\n", sum);
                break;
            case 2:
                printf("Row sums:\n");
                for (i = 0; i < r; i++) {
                    rowSum[i] = 0;
                    for (j = 0; j < c; j++)
                        rowSum[i] += a[i][j];
                    printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
                }
                printf("The row-sum array contains: [");
                for (i = 0; i < r; i++)
                    printf("%s%d", i ? ", " : "", rowSum[i]);
                printf("]\n");
                break;
            case 3:
                printf("Transpose of the matrix:\n");
                for (j = 0; j < c; j++) {
                    for (i = 0; i < r; i++)
                        printf("%d ", a[i][j]);
                    printf("\n");
                }
                break;
            case 4:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice.\n");
        }
    } while (ch != 4);
    return 0;
}
```

Sample Input and Output

```
Enter the number of rows and columns: 2 3
Enter the matrix elements:
1 2 3
4 5 6

1. Sum of all elements
2. Sum of each row
3. Transpose
4. Exit
Enter your choice: 1
Sum of all elements = 21

Enter your choice: 2
Row sums:
Sum of row 1 = 6
Sum of row 2 = 15
The row-sum array contains: [6, 15]

Enter your choice: 3
Transpose of the matrix:
1 4
2 5
3 6

Enter your choice: 4
Exiting program.
```

Result

The menu-driven C program was written successfully. It correctly calculated the sum of all matrix elements (21), stored the row sums ([6, 15]), and displayed the transpose of the given 2 x 3 matrix.